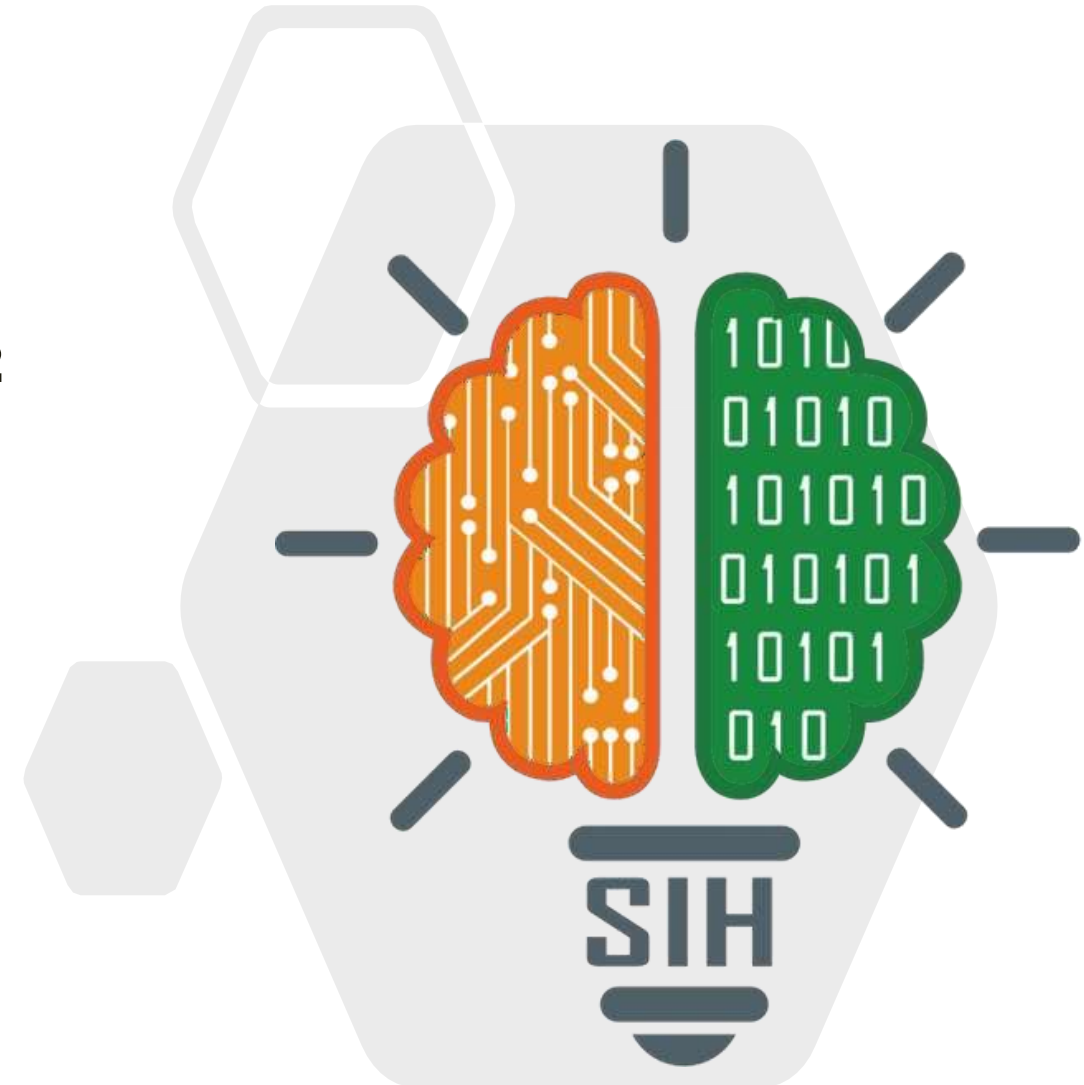


TITLE PAGE

- **Problem Statement ID - SIH26166**
- **Problem Statement Title –**
“Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS)”
- **Theme - Space Technology**
- **PS Category - Software**
- **Team ID - SIH26-033**
- **Team Name - Storm Breaker**





LUMINA -- AI-POWERED MULTI-MODAL LUNAR IMAGE CORRESPONDENCE



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PROPOSED SOLUTION

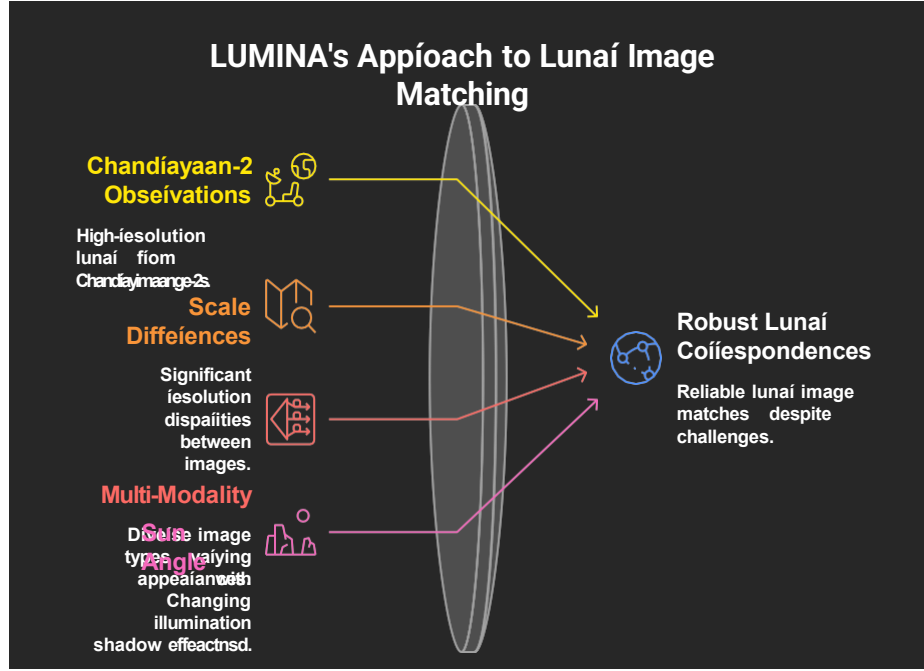
THE CORE CHALLENGES

SCALE DISPARITY : The same lunar surface appears at different scales because OHRC and TMC have different spatial resolutions. (OHRC 0.25 m/px ↔ TMC 5 m/px)

MULTI-MODAL OBSERVATION : OHRC, TMC and IIRS observe the lunar surface with different spatial and spectral characteristics (Panchromatic ↔ Spectral | Different spatial scales).

SUN-ANGLE VARIATION : Changing Sun angles alter shadows and surface appearance, making the same terrain look different. (South-Pole Sun elevation can be as low as ~1.5°)

RELIABLE CORRESPONDENCE : False or inaccurate matches can lead to incorrect lunar surface interpretation and mapping. (Geometric verification → RANSAC inlier validation)



WHY WE STAND OUT ?

Lumina : addresses the core challenges of lunar image correspondence by handling large scale differences between OHRC and TMC, variations in Sun-angle illumination through DEM-based ray tracing, and multi-modal observations from OHRC, TMC and IIRS—enabling reliable and geometrically verified correspondences across diverse lunar imagery.

Scale → Sun Angle → Multi-Modal → Reliable Correspondence

KEY FEATURES



LoFTR + RANSAC
Scale • Sun-Angle • Multi-Modal
78.4% Verified Inliers | 1.1 px Error

IMAGE MATCHING & GEOMETRIC VERIFICATION

DEM Ray-Tracing
Sun-Angle Normalization
1.5° Sun → 76 m
Shadow~35%
Matching Improvement



SHADOW ANALYSIS & DEM RAY TRACING



DFSAR Polarimetry +
Random Forest
CPR • DOP • L/S-band •
Thermal Inertia
94.0% Accuracy

DFSAR POLARIMETRY & ICE DETECTION

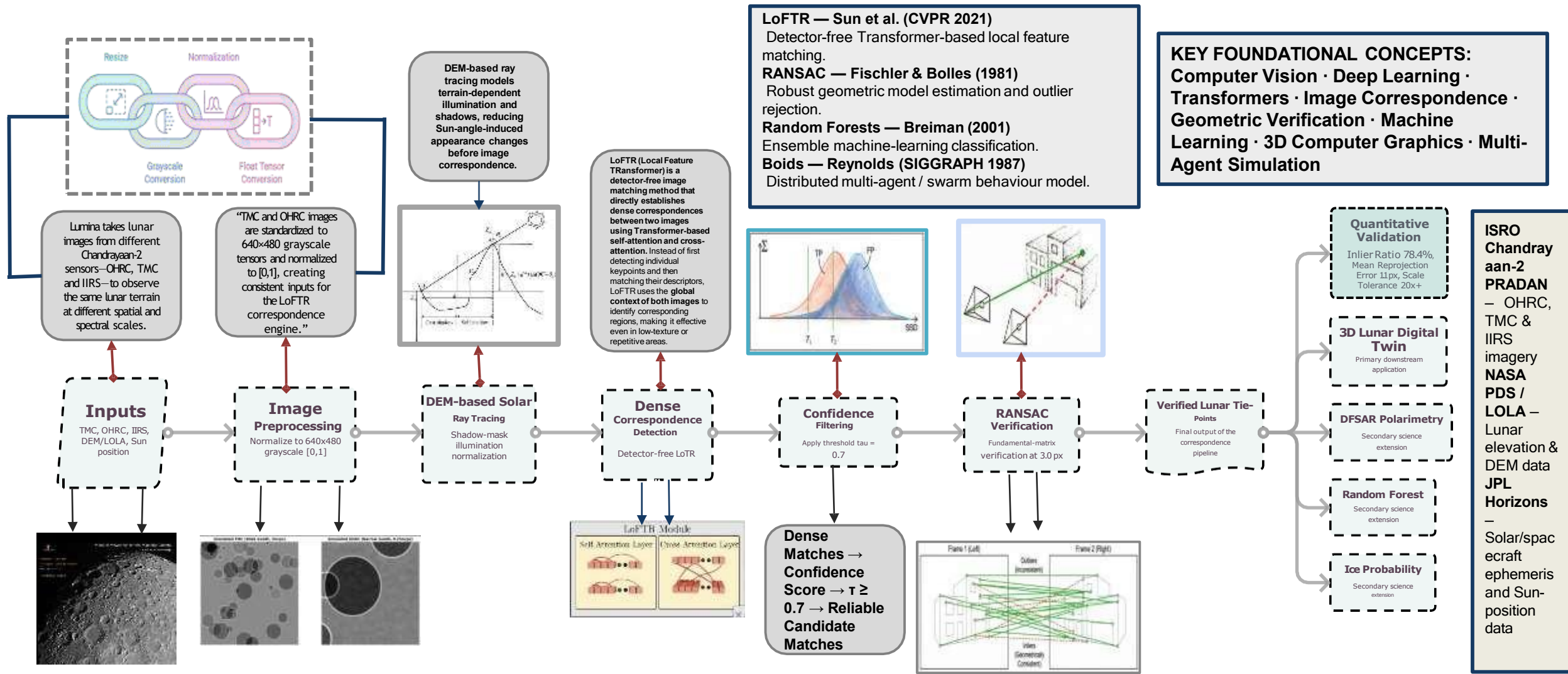


TECHNICAL APPROACH



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Lunar Multi-Modal Image Correspondence System





FEASIBILITY AND VIABILITY

POTENTIAL RISKS & CHALLENGES

1. · SCALE DISPARITY

OHRC-TMC images differ by ~20× in spatial scale.

2. · SUN-ANGLE VARIATION

Low Sun angles create strong shadows and appearance changes.

3. · MULTI-MODAL DIFFERENCES

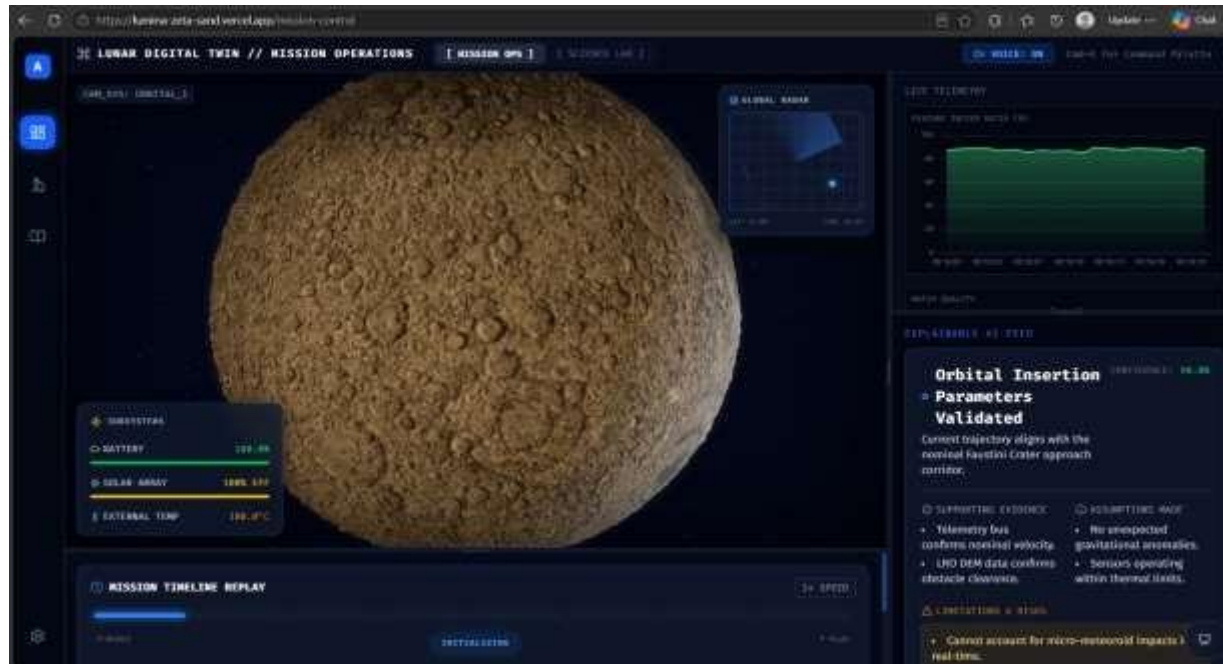
OHRC, TMC and IIRS provide different spatial and spectral information.

4. · COMPUTATIONAL LOAD

AI matching + 3D visualization can increase processing and GPU demand.

Bottom line:

Scale → Illumination → Modality → Computation



MITIGATION & VIABILITY

1. · SCALE

LoFTR + RANSAC → reliable geometric correspondences.

2. · SUN

DEM ray tracing → illumination-aware conditioning.

3. · MODALITY

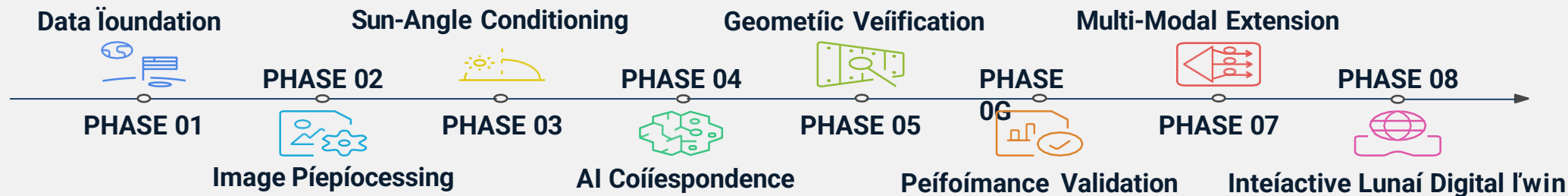
Common reference framework → scalable multimodal extension.

4. · COMPUTE

WebGL optimization → real-time browser visualization.

Bottom line:

Every major risk is paired with a technical mitigation and measurable evidence.



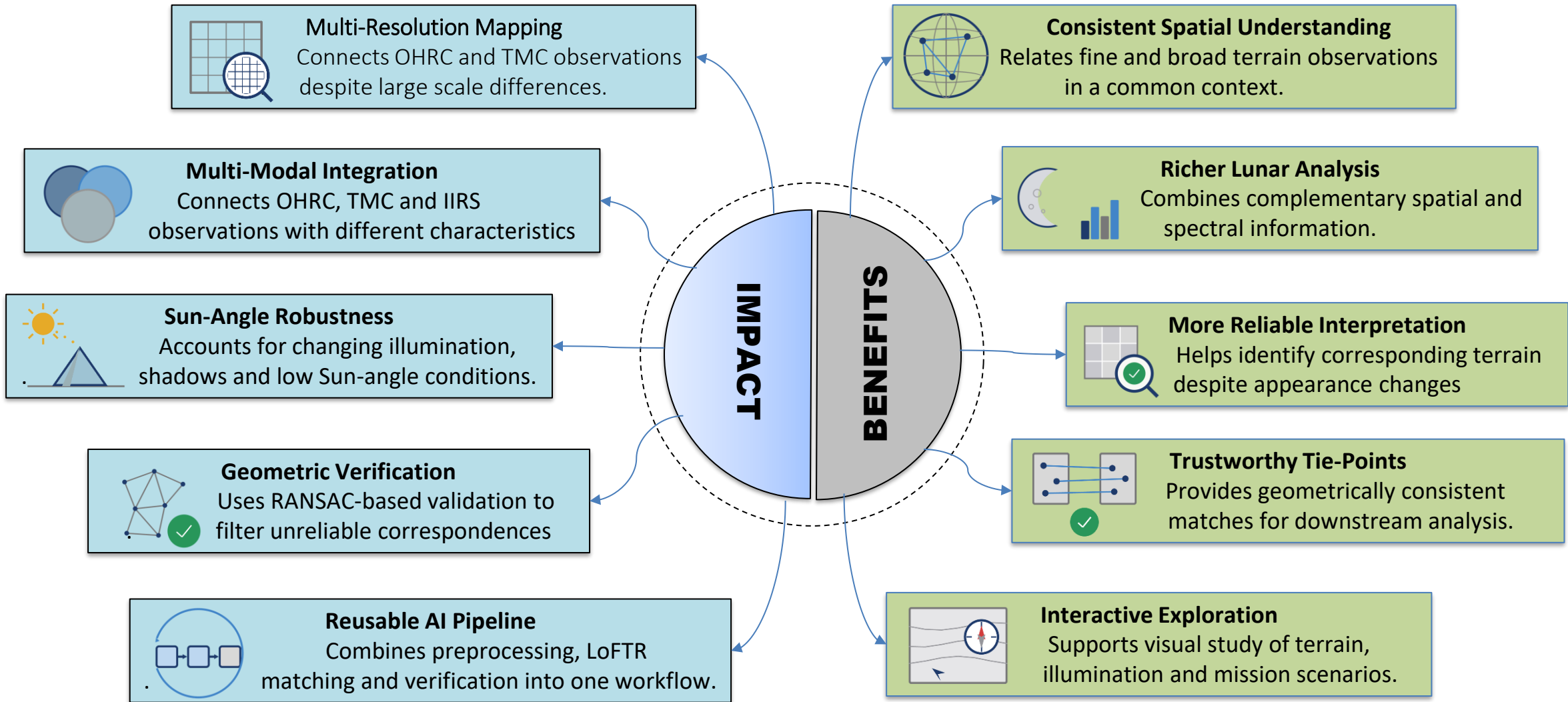
LUMINA v3 Project Development Timeline



IMPACT AND BENEFITS

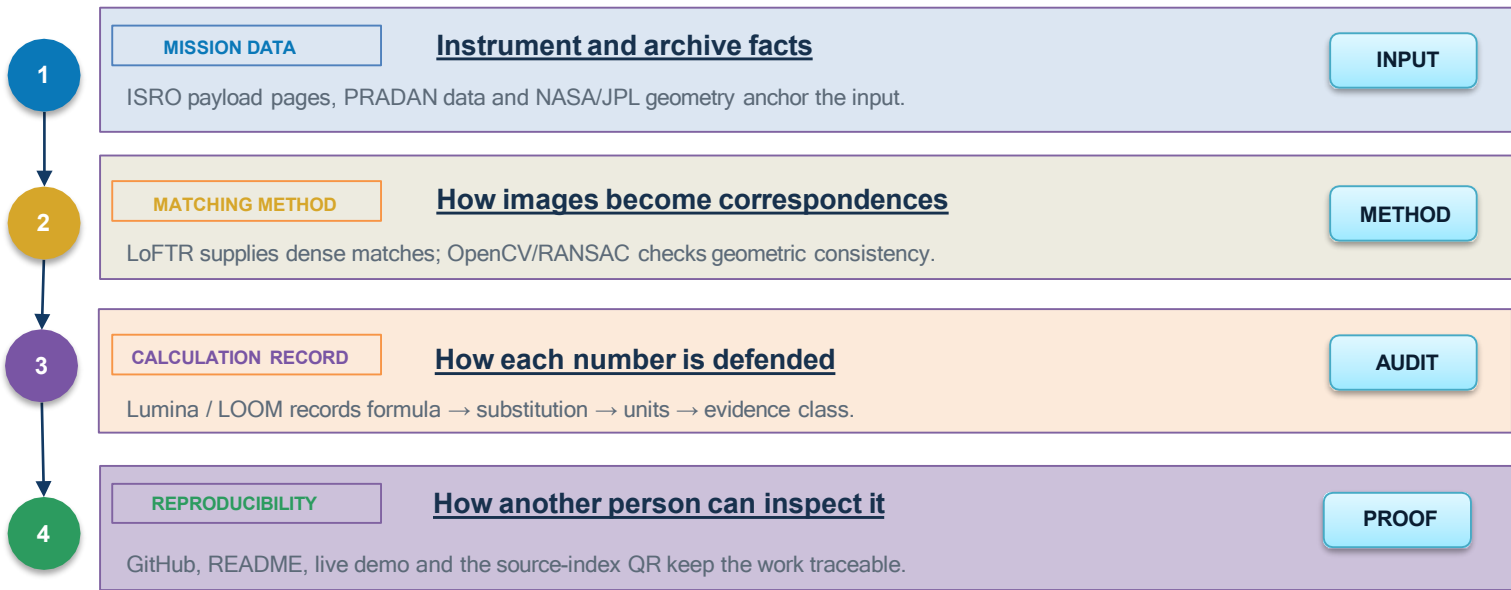


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HOW EACH SOURCE ENTERS THE PROJECT

What we took from the references — and what we computed ourselves



EVIDENCE LABELS USED IN THE AUDIT

- REFERENCE-DERIVED
- PROJECT CONFIG
- COMPUTED
- EXPERIMENTAL

OPEN PROOF



CODE + README
github.com/saichintamani/Lumina-



LIVE DEMO
lumina-zeta-sand.vercel.app



REFERENCE INDEX
all checked links

REFERENCE REGISTER

Where each method, fact and link came from

SOURCE	USED FOR	SHORT LINK
ISRO payloads	TMC-2 / OHRC / IIRS / DFSAR specs	isro.gov.in
NASA / JPL	Lunar geometry and timing	ssd.jpl.nasa.gov
PRADAN / ISSDC	Public lunar data archive	pradan.issdc.gov.in
OpenCV calib3d	RANSAC / geometric checks	docs.opencv.org
LoFTR · CVPR 2021	Detector-free image matching	CVPR paper
Reynolds · 1987	Multi-agent steering model	red3d.com
SIFT · Lowe	Sparse-feature baseline	Cornell PDF
Metrics / RF	Classification and forest metrics	Google ML / sklearn
Lumina records	Formula, units and benchmarks	project record
GitHub + Vercel	Code, README and live proof	repo / demo
Reference video	No video URL in supplied packet	not supplied